

# SYSTEM REPORT

## Dynamic Modelling and Simulation of a Conveyor Belt Under Variable Load Conditions

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## 1. System Selection and Approval

The selected system is a conveyor belt transporting a block under Coulomb friction, categorized under Category B. The system was chosen due to its relevance in industrial automation and its suitability for applying dynamic system modelling concepts such as Newton's laws, friction modelling, and simulation validation.

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## 2. Real-World Application

Conveyor belt systems are widely used in manufacturing, mining, and airport baggage handling. In these systems, loads are often placed on moving belts, leading to slip, friction-driven acceleration, and synchronization challenges. Understanding this behaviour is essential for improving efficiency, reducing material loss, and optimizing system performance.

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## 3. System Definition

### 3.1 Inputs

- Belt velocity,  $V_{\text{belt}} = 1.5 \text{ m/s}$  ( $V_{\text{belt}} = 1.5 \text{ m/s}$  (constant))
- Coefficient of friction,  $\mu$

### 3.2 Outputs

- Block velocity,  $v(t)$
- Block position,  $x(t)$
- Kinetic energy,  $KE(t)$

### 3.3 State Variables

$$X = [v, x]^T \quad \dot{X} = [a, v]^T$$

### 3.4 Key Parameters

- Mass,  $m = 1.0 \text{ kg}$
- Friction coefficient,  $\mu = 0.35$
- Gravity,  $g = 9.81 \text{ m/s}^2$
- Belt velocity,  $V_{\text{belt}} = 1.5 \text{ m/s}$

## 4. Degrees of Freedom (DOF)

The system has **1 degree of freedom**, representing translational motion along the belt. No rotational or vertical motion is considered, simplifying the system to a point mass model.

## 5. Free Body Diagram (FBD)

The block is subjected to:

- Weight  $mg$  (downward)
- Normal reaction  $N = mg$  (upward)
- Friction force:

$$F_f = \mu mg \cdot \text{sign}(V_{\text{belt}} - v)$$

When the block velocity is less than belt velocity, friction accelerates the block. Once both velocities match, friction becomes zero.

## 6. Mathematical Modelling

Using Newton's Second Law:

$$m \frac{dv}{dt} = \mu mg \cdot \text{sign}(V_{\text{belt}} - v)$$

Simplified:

$$dv/dt = \mu g \cdot \text{sign}(V_{\text{belt}} - v) \quad dt dv = \mu g \cdot \text{sign}(V_{\text{belt}} - v)$$

Kinematic equation:

$$dx/dt = v \quad dt dx = v$$

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## 7. State-Space Representation

State vector:

$$X = [v, x]^T \quad X = [v, x]^T$$

Equations:

$$\dot{x}_1 = \mu g \cdot \text{sign}(V_{\text{belt}} - x_1) \quad \dot{x}_1 = \mu g \cdot \text{sign}(V_{\text{belt}} - x_1) \quad \dot{x}_2 = x_1 \quad \dot{x}_2 = x_1$$

The system is nonlinear due to the sign function.

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## 8. Energy Analysis

Kinetic energy is given by:

$$KE = \frac{1}{2}mv^2 \quad KE = \frac{1}{2}mv^2$$

At steady state:

$$KE = 1.125 \text{ J} \quad KE = 1.125 \text{ J}$$

Friction performs work on the block, increasing its energy until it reaches belt speed.

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## 9. Simulation Details

- Software: Python (NumPy, SciPy, Matplotlib)
- Solver: `solve_ivp` (RK45 method)
- Initial conditions:  $v(0)=0$ ,  $x(0)=0$
- Simulation time: 6.5 seconds

The block reaches the belt velocity of 1.5 m/s at approximately 0.43 seconds.

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## 10. Results and Discussion

- Velocity increases linearly due to constant acceleration
  - Position increases continuously over time
  - After reaching belt speed, velocity remains constant
  - Phase behaviour confirms constant acceleration
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## 11. Validation

### Energy Validation:

Steady-state kinetic energy matches analytical value (1.125 J) ✓

### Analytical Validation:

$$t = V_{\text{belt}} \mu g = 0.437 \text{ s} \quad t = \mu g V_{\text{belt}} = 0.437 \text{ s}$$

Simulation:  $t \approx 0.43 \text{ s}$   $t \approx 0.43 \text{ s}$  ✓

### Dimensional Check:

$$dv/dt = \mu g \quad dt/dv = \mu g$$

Units are consistent ✓

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## 12. Sensitivity Analysis

| $\mu$ | Time to reach belt speed<br>(s) |
|-------|---------------------------------|
| 0.2   | 0.76                            |
| 0.35  | 0.43                            |
| 0.5   | 0.31                            |

Higher friction leads to faster acceleration and quicker synchronization.

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## 13. Assumptions

- Point mass model (no rotation)
  - Constant belt velocity
  - Coulomb friction only
  - No air resistance
  - Rigid surfaces
  - No external disturbances
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## 14. Limitations

- Belt elasticity not considered
  - Motor dynamics ignored
  - No experimental validation
  - No vibration effects included
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## 15. Conclusion

The model successfully captures the dynamic behaviour of a block placed on a moving conveyor belt. The block accelerates uniformly due to Coulomb friction and reaches belt speed in approximately 0.43 seconds. The simulation results closely match analytical predictions, validating the model.

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## 16. References

- Ginsberg, J. *Engineering Dynamics*, Cambridge University Press
- Beer & Johnston, *Vector Mechanics for Engineers*
- SciPy Documentation – `solve_ivp`
- Matplotlib Documentation